



Fig. 5: Encrypted payload format in counter with cipher block chaining-MAC mode of AES

icates the subscriber station, it may protect against an attacker employing a cloned subscriber station masquerading as a legitimate subscriber.

Privacy key management RSA authentication. The PKM RSA authentication protocol uses X.509 digital certificates—the RSA public key encryption algorithm that binds public RSA encryption keys to MAC addresses of subscriber stations. The digital certificate contains the subscriber station's public key and MAC address. When requesting an authentication key, a subscriber station presents its digital certificate to the base station. The base station verifies the digital certificate and then uses the verified public key to encrypt an authentication key, which the base station then sends back to the requesting subscriber station (refer Fig. 3).

All subscriber stations using RSA authentication have factory-installed RSA private/public key pairs or provide an internal algorithm to generate such key pairs dynamically. If a subscriber station relies on an internal algorithm to generate its RSA key pair, it generates the key pair prior to its first authorisation key exchange. All subscriber stations with factory-installed RSA key pairs also have factory-installed X.509 certificates.

PKM extensible authentication protocol authentication. This authentication uses extensible authentication protocol (EAP) in conjunction with an operator-selected EAP method (e.g., EAP-transport layer security (TLS) or EAP-tunnelled TLS with Microsoft challenge handshake authentication protocol version 2 (TTLS MS-CHAPv2)). The method uses a par-

ticular kind of credential—X.509 certificate in case of EAP-TLS, or a subscriber identity module in case of EAP-SIM (refer Fig. 4).

EAP authentication provides dynamic encryption keys to wireless users. These are more secure than static encryption keys. If an intruder passively receives enough packets encrypted by the same encryption key, he can perform a calculation to learn the key

and use it to join a network. Because dynamic encryption keys change frequently, these prevent intruders from performing the calculation and learning the key.

After successful EAP-based authentication, if the subscriber station or base station negotiates authorisation policy as 'authenticated EAP after EAP' mode, the authenticated EAP messages carries the second EAP message. It cryptographically binds previous and following EAP authentication sessions, while protecting second EAP messages. In order to prevent 'man-in-the-middle attack', the first and second EAP methods should fulfil the mandatory criteria.

During re-authentication, the EAP transfer messages are protected with a hashed message authentication code or cipher message authentication code protocol. The base station and subscriber station discard unprotected EAP transfer messages or EAP transfer messages with invalid hashed message authentication code or cipher message authentication code digests during re-authentication.

Message authentication code keys are used to sign management messages in order to validate the authenticity of these messages. The message authentication code to be used is negotiated at a subscriber station's basic capabilities negotiation request. There are different keys for uplink and downlink messages. For a multicast message (in downlink only) and for a unicast message, a different message authentication key is generated. In general, message authentication keys used to generate the cipher message authentication code value and the hashed message authentication code-digest are derived from the authorisation key.

Cryptographic method of data encryption

Encryption services are defined as a set of capabilities within the MAC security sub-layer. Encryption information is allocated in the generic MAC header format and applied to the MAC packet data unit payload when required by the selected ciphersuite while the generic MAC header is not encrypted.

With advanced encryption standard's counter with cipher block chaining-MAC mode (an authenticated encryption algorithm designed to provide both authentication and confidentiality), the packet data unit payload (after encryption) has a format as shown in Fig. 5. Here packet data unit payload is prepended with a 4-byte packet number. The packet number is transmitted with the least significant bit first and the packet number itself is not encrypted. The packet number associated with a security association is set to '1' when the security association is established and a new traffic encryption key installed. After each packet data unit transmission, the packet number is incremented by '1'.

On uplink connections, the packet number is XORed with 0x80000000 prior to encryption and transmission. On downlink connections, the packet number is used without such modification. The ciphertext message authentication code is transmitted such that byte index '0' is transmitted first and byte index '7' transmitted last.

The result

WiMAX users should feel confident that their transmitted data is free from eavesdropping or manipulation and only authorised users can access WiMAX services. The improved WiMAX technology implements improved security architecture. Thus the operator ensures that only authorised subscribers access the network and use appropriate services subscribed by them. ●

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